

TDV 2215 Complete NVRAM Memory Map

Overview

The TDV 2215 uses an ER3400 series non-volatile RAM (NVRAM) to store configuration data that persists across power cycles. This document provides a complete mapping of the NVRAM usage based on firmware analysis and the official specification.

NVRAM Architecture

- **Chip:** ER3400 series NVRAM
- **Addressing:** CPU addresses mapped via `map_to_nvram_address()` function
- **Control:** A10 bit controls ER3400 mode (read/write vs erase/accept_address)
- **Access:** /S1 signal from R/W controls C0 (read vs write operations)

Complete NVRAM Memory Layout

Soft-Switch Configuration Data (0x50-0x5A) - 11 bytes

Packed bit-field storage for all 37 soft-switches, extracted via template system:

NVRAM Address	Content	Template Byte	Switches Stored
0x50	Config Byte 0	0x0AA0 (0xE8)	Convenience: Cursor Type, Key Click, Margin Bell
0x51	Config Byte 1	0x0AA1 (0xE8)	Convenience: Auto Repeat, Key Rollover, Keyboard CAPS
0x52	Config Byte 2	0x0AA2 (0xFF)	Function: Time-out, Bell, Graphic Rendition Mode
0x53	Config Byte 3	0x0AA3 (0xF8)	Function: Underline Rep, Cursor Return, BOL Wrap
0x54	Config Byte 4	0x0AA4 (0xAF)	Function: EOL Wrap, Roll/Page Mode, Roll Type
0x55	Config Byte 5	0x0AA5 (0x80)	Function: Clear Lamps, PUSH-key Prog, Extended Control
0x56	Config Byte 6	0x0AA6 (0x80)	Function: Vertical Edit, Printer Mode, Printer FF
0x57	Config Byte 7	0x0AA7 (0x80)	Communication: SRM, Echo, Line/Local Status
0x58	Config Byte 8	0x0AA8 (0x80)	Communication: Online, Comm Clock, Handshake
0x59	Config Byte 9	0x0AA9 (0x80)	Communication: Modem, Code Length, Parity
0x5A	Config Byte 10	0x0AAA (0x80)	Communication: Stop Bits, Speed, Delay, Mode, Printer settings

PUSH-Key Sequence Storage (16 sequences, 12 bytes each)

8 physical keys × 2 modes (normal/shifted) = 16 programmable sequences:

Key	Normal Mode	Shifted Mode	Description
PUSH-Key 1	0x60-0x6B	0x6C-0x77	12 bytes each
PUSH-Key 2	0x78-0x83	0x84-0x8F	12 bytes each
PUSH-Key 3	0x90-0x9B	0x9C-0xA7	12 bytes each
PUSH-Key 4	0xA8-0xB3	0xB4-0xBF	12 bytes each
PUSH-Key 5	0xC0-0xCB	0xCC-0xD7	12 bytes each
PUSH-Key 6	0xD8-0xE3	0xE0-0xEB	12 bytes each
PUSH-Key 7	0x100-0x10B	0x120-0x12B	12 bytes each
PUSH-Key 8	0x140-0x14B	0x170-0x17B	12 bytes each

Total PUSH-Key Storage: 192 bytes (16 × 12 bytes)

Tabulation Rack Storage

Location and size determined by configuration menu system:

- Horizontal tab stop positions
- Permanent/temporary tab settings
- Menu-controlled save/restore ("Make Tabulation Rack Permanent")

Configuration Processing

Template System

1. **Template Table:** Located at 0x0AA0-0x0AAA in ROM (11 bytes)
2. **Bit Extraction:** Each template byte defines bit-field extraction pattern
3. **Output:** 37 individual switch values stored at RAM 0x5F00-0x5F24
4. **Process:** `load_config_from_nvram_template()` function

Switch Categories and Default Values

Convenience Switches (6 switches) - Section 4.1

1. **Cursor Type:** LINE (default), BLOCK
2. **Key Click:** ON (default), OFF
3. **Margin Bell:** ON (default), OFF
4. **Auto Repeat:** ON (default), OFF
5. **Key Rollover:** ENABLED (default), DISABLED
6. **Keyboard CAPS on Power Up:** OFF (default), ON

Function Switches (15 switches) - Section 4.2

7. **Time-out:** ON (default), OFF
8. **Bell:** ON (default), OFF
9. **Graphic Rendition Mode:** ATTR (default), UNDERLINE, SGR
10. **Underline Representation:** UNDERLINE (default), multiple options
11. **Cursor Return:** CR (default), CRLF
12. **Beginning of Line Wrap:** STOP (default), WRAP
13. **End of Line Wrap:** STOP (default), WRAP
14. **Roll/Page Mode:** ROLL (default), PAGE
15. **Roll Type:** STEP (default), SMOOTH
16. **Clear Lamps:** BOTH (default), KEY, SYN
17. **PUSH-key Programming:** ALLOWED (default), PROHIBITED
18. **Extended Control:** OFF (default), ON - **CRITICAL COMPATIBILITY SWITCH**
19. **Vertical Editing Mode:** FOLLOWING (default)
20. **Printer Mode:** LOCAL/REM (default), REMOTE, LOG
21. **Printer Form Feed:** NONE (default), BEFORE, AFTER, BOTH

Communication Switches (16 switches) - Section 4.3

22. **Send Receive Mode:** SIMULTANEOUS (default), TRANSPARENT
23. **Echo:** ON (default), OFF
24. **Line/Local Status:** LINE (default), LOCAL
25. **Online:** ONLINE (default), TOGGLE
26. **Communication Clock:** ASY (default)
27. **Communication Handshake:** OFF (default), XON/XOFF, DTR
28. **Modem:** INHIBIT (default), LEASED, DIALLED
29. **Transmission Code Length:** 7BIT (default), 8BIT
30. **Transmission Code Parity:** NONE (default), EVEN, ODD
31. **Transmission Code Stop Bits:** 1BIT (default), 2BIT
32. **Transmission Speed:** 9600 baud (default), 50-19200 range
33. **Transmission Delay:** NONE (default), 20ms, 40ms, 60ms
34. **Communication Mode:** V.24 (default), V.11
35. **Printer Handshake:** OFF (default), XON/XOFF
36. **Printer Code Format:** 7EVEN (default), 7ODD, 8NONE, 8EVEN, 8ODD
37. **Printer Speed:** 1200 baud (default), 50-19200 range

NVRAM Management

Power-Up Sequence

1. Hardware power-on test includes NVRAM validation
2. Load permanent switches from NVRAM → temporary switches in RAM
3. Apply default values if NVRAM uninitialized or corrupted
4. Initialize terminal based on loaded configuration

Configuration Menu System (Section 5)

- **Access:** MODE key twice while SHIFT pressed
- **Categories:** Convenience, Function, Communication switches + PUSH-keys + Tabulation
- **Permanent/Temporary:** Changes affect RAM copies, menu controls NVRAM write-back
- **Factory Reset:** Restore all defaults and save to NVRAM

Remote Configuration

- **Host Control:** Most switches controllable via SM/RM commands (except communication format switches)
- **PUSH-Key Programming:** DCS (Device Control String) sequences from host
- **Limitations:** Host cannot access setup menus or make settings permanent

Technical Notes

NVRAM Reliability

- ER3400 series provides data retention without power
- Memory test functions validate NVRAM integrity during power-up
- Error codes displayed if NVRAM failures detected

Compatibility

- **TDV 2115 Mode:** Extended Control = OFF, limited to C0 control codes
- **Extended Mode:** Extended Control = ON, full TDV 2215 capabilities
- **Activation:** ESC Q sequence enables extended mode from TDV 2115 mode

Programming Interface

- **Local:** Setup menus (if PUSH-key Programming = ALLOWED)
- **Remote:** DCS sequences (always available when Extended Control = ON)
- **Format:** Text strings and/or control character sequences
- **Length:** Maximum 12 bytes per PUSH-key sequence

This comprehensive NVRAM map ensures proper understanding of the TDV 2215's non-volatile storage system and configuration management.